

THE FUNDAMENTAL UNIT OF LIFE

Cell

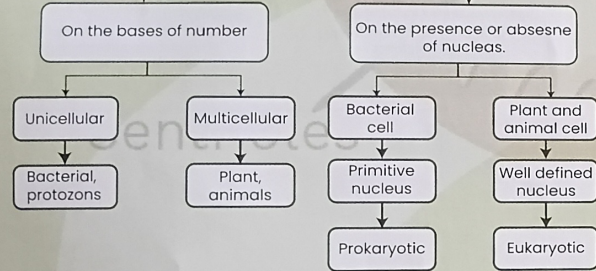
1 Discovery

Robert Hooke in 1665
Robert Brown discovered nucleus in the cell in 1831
Anton Van Leeuwenhoek observed the living cell in pond water with the help of primitive microscope in 1674.

2 Cell theory

Schleiden and Schwann gave cell theory which states that all plants and animals are composed of cells.
Rudolf Virchow (1855) modified the cell theory and explained that the cells are divided to form new cells from pre-existing cells. (*Omnis cellula-e cellula*).

3 Classification of cells



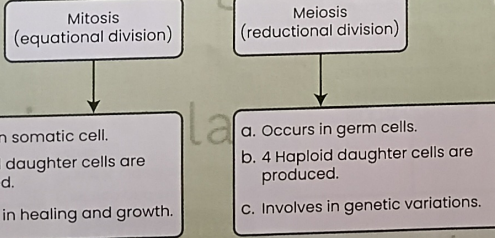
4 Characteristics of plant cell

- Cell wall present
- Centrosome absent
- Chloroplast present
- Vacuole larger in size

5 Characteristics of animal cell

- Lack cell wall
- Centrosome present
- Chloroplast absent
- Vacuole are absent or smaller in size

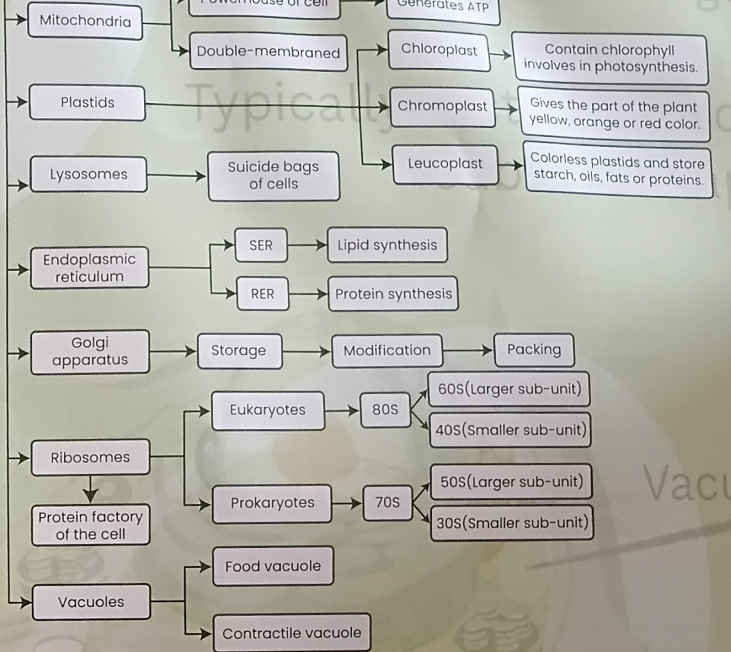
6 Division



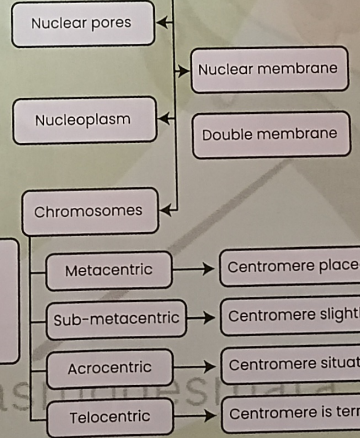
9 Cytoplasm

- 2. Made up of bilayer of phospholipids and proteins.
- 1. Selectively permeable

10 Organelles



8 Nucleus



Transport of substances

